

Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

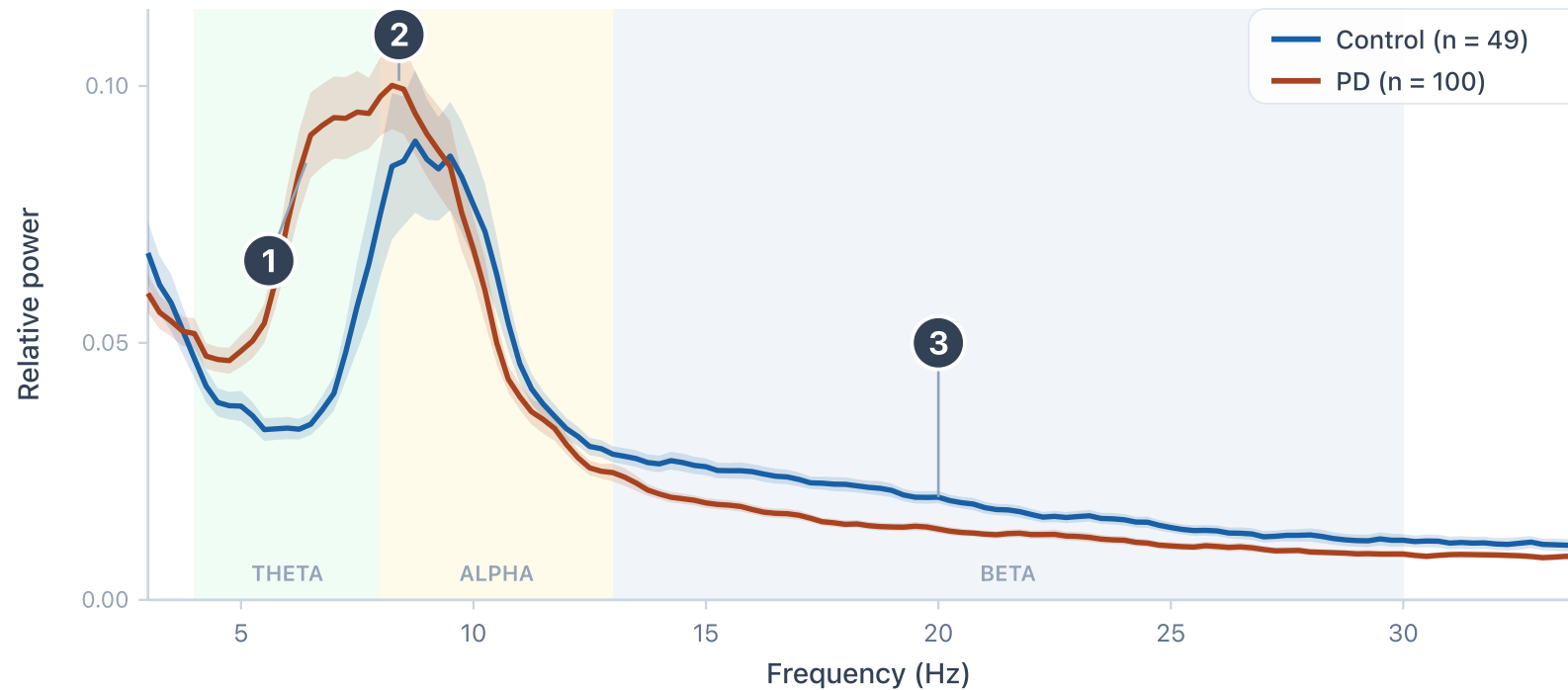
PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

Hypotheses

① theta power higher
in PD · $p < 0.001$

② alpha peak slows
9.25 → 8.00 Hz · $p < 0.001$

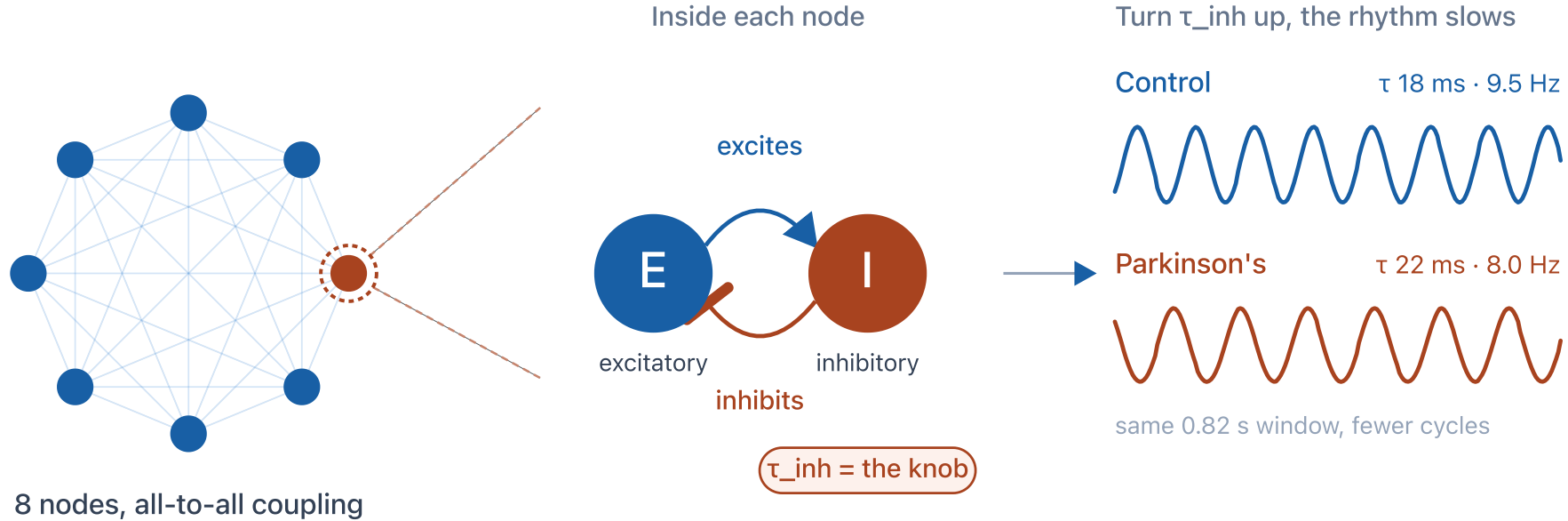
③ beta power lower
in PD · $p < 0.001$



Alpha power itself is unchanged: the slowing is in frequency, not amplitude.

The model: a Wilson-Cowan network

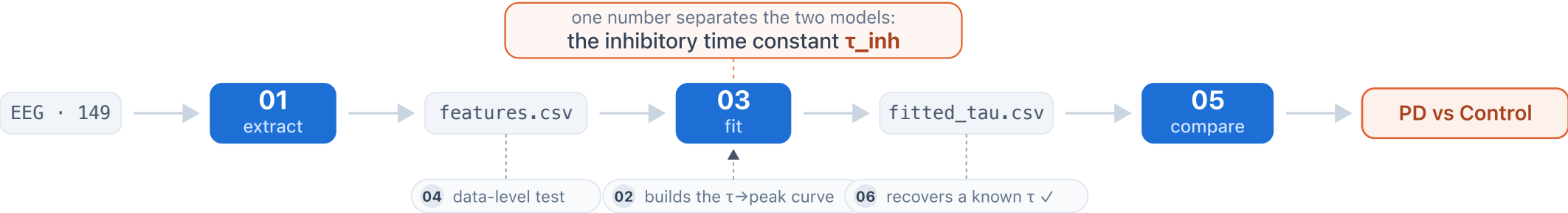
8 coupled cortical nodes, each an excitatory-inhibitory loop. τ_{inh} is the only knob we turn.



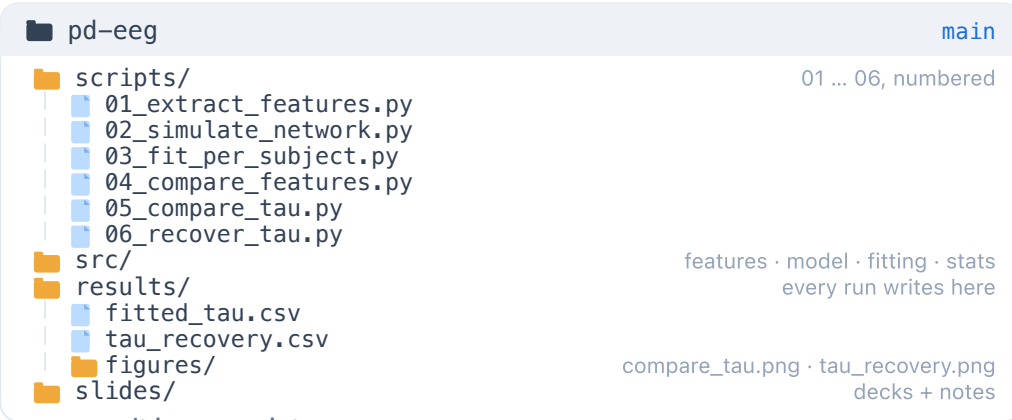
The final adaptation from healthy to patient is a single parameter: a longer inhibitory time constant τ_{inh} , about **+3.25 ms** in PD.

Code & repository

THE PIPELINE EEG to result, through six numbered scripts



THE REPOSITORY

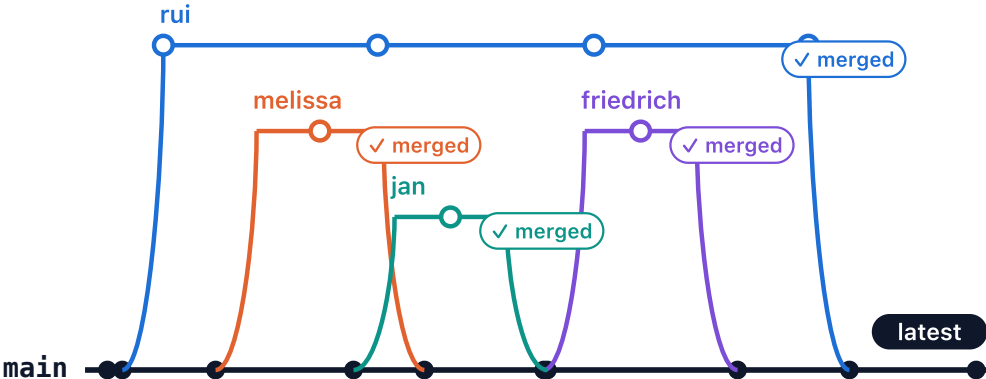


every result is one script run

```
bash$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

feature branches, reviewed, merged into main



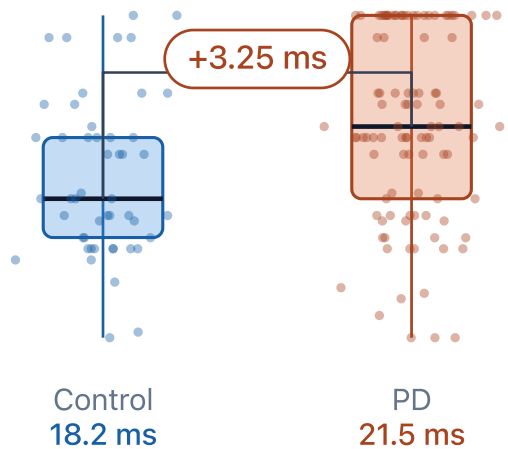
50 commits · feature branches → reviewed → merged into main · fully reproducible

Does the model separate patients from controls?

Fitted inhibitory time constant

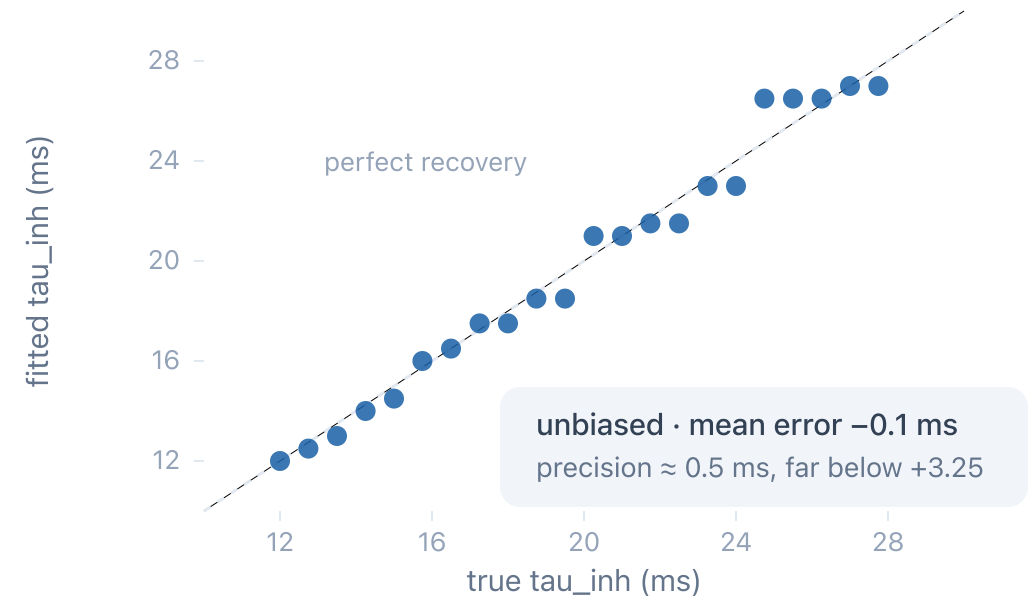
the model parameter we change for a patient

fitting recovers tau to ± 0.5 ms, far below this group gap



Validation: recovering a known parameter

simulate at a known tau, fit it back



Fitted τ_{inh} is **+3.25 ms** higher in PD (one-sided Mann-Whitney, $p \approx 0.0002$). The recovery test shows this is a real separation, not an artefact of the fitting.

Outlook and open challenges

CURRENT CHALLENGES

- One parameter (τ_{inh}) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted τ to clinical scores and to known inhibitory (GABAergic) changes in PD

What we took away

Rui [your biggest learning]

Jan [your biggest learning]

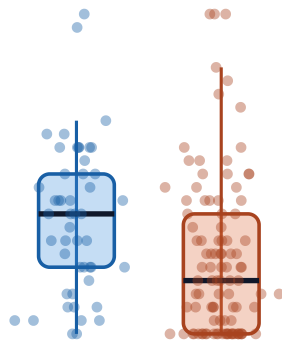
Melissa [your biggest learning]

Friedrich [your biggest learning]

Appendix: per-subject distributions

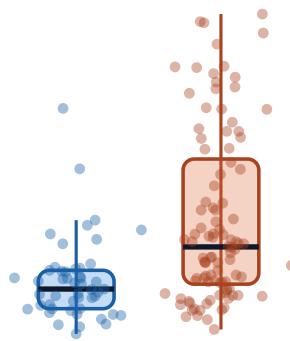
PD vs Control: per-subject EEG features

Alpha peak frequency
slower in PD



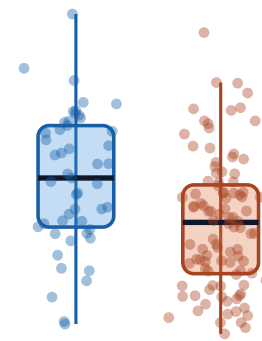
Control PD

Theta power
higher in PD



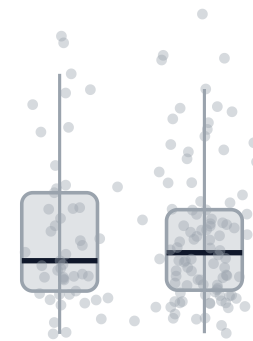
Control PD

Beta power
lower in PD



Control PD

Alpha power
unchanged



Control PD

Appendix: effect sizes

How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval

